

Network Troubleshooting (Easy English Notes)

1. What is Network Troubleshooting?

Imagine the electricity or water suddenly stops working in your house.

Then you start checking:

- Where is the problem?
- Is the switch OFF?
- Is the wire disconnected?
- Is there a bill/payment issue?

Finding and fixing problems in a network is called **Network Troubleshooting**.

Meaning: Finding the problem and fixing it.

Real Example

Problem: Internet is not working

You check:

- Is Wi-Fi connected?
- Does the device have an IP address?
- Is the router running?
- Is there an ISP issue?

This is called troubleshooting.

2. Common Network Problems

1. No Connectivity

Possible reasons:

- Cable disconnected
- Wi-Fi turned OFF
- Wrong IP address

Example:

PC ■ Router

2. Slow Network

In networking:

- Too many users
- Heavy YouTube streaming/downloads
- Overloaded router

Internet becomes slow.

3. IP Conflict

Example:

PC1 = 192.168.1.10

PC2 = 192.168.1.10

This creates an IP conflict and causes problems.

4. DNS Problem

DNS converts:

Website Name → **IP Address**

Example:

google.com → 142.xx.xx.xx

If DNS fails:

- Website won't open by name
- But it may still work using the IP address

5. Firewall Block

If firewall rules are wrong, websites or apps may be blocked.

3. Troubleshooting Steps

Step 1: Identify the Problem

No internet? Slow network? Website not opening?

Step 2: Gather Information

Check Wi-Fi, IP address, router lights.

Step 3: Isolate the Problem

PC? Cable? Switch? Router? ISP?

Step 4: Fix and Test

Restart the router or change DNS settings.

Step 5: Verify

Check if the internet works now.

4. Important Commands

ipconfig

Command: ipconfig /all

Shows IP Address, Default Gateway, DNS Server

ping

Command: ping 8.8.8.8

If reply comes → Connection OK

No reply → Problem exists

tracert (Traceroute)

Command: tracert google.com

Shows path: PC → Router → ISP → Google

nslookup

Command: nslookup google.com

Finds website IP address

netstat

Command: netstat -an
Shows active/open connections

Key Terms

Word	Meaning
Ping	Is the device alive?
Traceroute	Which path did data take?
DNS	Website Name → IP
Bandwidth	How wide the road is
Latency	How long data takes to travel

Real Technician Trick

1. Check Cable/WiFi
2. Restart Router
3. Run ipconfig
4. Ping Gateway
5. Ping 8.8.8.8
6. Ping google.com

Where it fails = the problem is likely there.